# PROPERNESS APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 2033-2070, following the statement at 2022-2031.

*Editorial note.* No omissions; Situation situation-descent-property supplies f = lim f_i with f_i of finite type.

*Allowed standard prerequisites (editorial).*

- Proper is etale-local on the base. Morphisms of Spaces, Lemma spaces-morphisms-lemma-proper-local (cross-chapter, cited at ~2036-2038). Reduces to the affine etale base V_0 -> Y_0.
- Descent of separated morphisms. lemma-descend-separated-morphism (same chapter, cited at ~2042-2044). f_i is separated for some i.
- Weak Chow's lemma for algebraic spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-weak-chow (cross-chapter, cited at ~2046-2048). Diagram with X_0' -> P^n_{Y_0} an immersion and pi : X_0' -> X_0 proper surjective.
- Properness permanence. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-composition-proper and spaces-morphisms-lemma-base-change-proper (cross-chapter, cited at ~2058-2061). X' -> Y is proper.
- Proper plus immersion is a closed immersion. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-universally-closed-permanence (cross-chapter, cited at ~2061-2063). X' -> P^n_Y is a closed immersion.
- Image of a proper morphism is proper. Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-proper-is-proper (cross-chapter, cited at ~2064-2065). It suffices to show X_i' -> Y_i proper for some i.
- Descent of closed immersions. lemma-descend-closed-immersion (same chapter, cited at ~2066-2068). X_i' -> P^n_{Y_i} is a closed immersion for i large.

*Source reference.* Original lemma 0850: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1887–1887. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0851: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1911–1911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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*Date*: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 089J: focused-seeds/geometry/S9-spaces-limits-eventually-proper/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04WP: focused-seeds/geometry/S9-spaces-limits-eventually-proper/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04XY: focused-seeds/geometry/S9-spaces-limits-eventually-proper/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 08AJ: focused-seeds/geometry/S9-spaces-limits-eventually-proper/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 083R: focused-seeds/geometry/S9-spaces-limits-eventually-proper/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04NX: focused-seeds/geometry/S9-spaces-limits-eventually-proper/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Assumptions and notation as in Situation 084W. If*

(1) *$f$ is proper, and*
(2) *$f_0$ is locally of finite type,*

*then there exists an $i$ such that $f_i$ is proper.*

*Proof.* Choose an affine scheme $V_0$ and a surjective étale morphism $V_0 \to Y_0$. Set $V_i = Y_i \times_{Y_0} V_0$ and $V = Y \times_{Y_0} V_0$. It suffices to prove that the base change of $f_i$ to $V_i$ is proper, see Morphisms of Spaces, Lemma 083R. Thus we may assume $Y_0$ is affine.

By Lemma 0851 we see that $f_i$ is separated for some $i \geq 0$. Replacing 0 by $i$ we may assume that $f_0$ is separated. Observe that $f_0$ is quasi-compact. Thus $f_0$ is separated and of finite type. By Cohomology of Spaces, Lemma 089J we can choose a diagram

$$\begin{array}{ccccc} X_0 & \xleftarrow{\pi} & X_0' & \longrightarrow & \mathbf{P}^n_{Y_0} \\ & \searrow & \downarrow & \swarrow & \\ & & Y_0 & & \end{array}$$

where $X_0' \to \mathbf{P}^n_{Y_0}$ is an immersion, and $\pi : X_0' \to X_0$ is proper and surjective. Introduce $X' = X_0' \times_{Y_0} Y$ and $X_i' = X_0' \times_{Y_0} Y_i$. By Morphisms of Spaces, Lemmas 04XY and 04WP we see that $X' \to Y$ is proper. Hence $X' \to \mathbf{P}^n_Y$ is a closed immersion (Morphisms of Spaces, Lemma 04NX). By Morphisms of Spaces, Lemma 08AJ it suffices to prove that $X_i' \to Y_i$ is proper for some $i$. By Lemma 0850 we find that $X_i' \to \mathbf{P}^n_{Y_i}$ is a closed immersion for $i$ large enough. Then $X_i' \to Y_i$ is proper and we win. $\square$